

Homework 1: One-Sample Inference

Lions like to eat onions

1 Question 1

1.1 Part a

The data are the proportions of successful retrievals out of 12 tests for each dog, so they are quantitative rather than simple 0/1 Bernoulli trials. Thus, a one-sample z-test is not a good fit.

A one-sample t-test and Wilcoxon signed-rank test are plausible.

For the one-sample t-test, the population does not appear to be approximately Gaussian, since there are quite a few observations at the upper bound of 1 and the data are clearly left-skewed. The sample size ($n = 10$) is not large enough to rely on the CLT either. For the Wilcoxon signed-rank test, the data do not appear symmetric for the same reason stated above. The sample size is not large enough for the asymptotic approximation either.

One noticeable thing is that quite a few observations have the same value, which introduces ties if the Wilcoxon test is applied. Furthermore, there are several zeros in the second data set (paired difference), which complicates the ranking process. Thus, I might choose to use the one-sample t-test.

1.2 Part b

1.2.1 Research Question and Data

Statistical Questions:

1. Whether the dogs were better than chance, which we assume is 50%, at returning the new objects when requested.
2. Whether there was a difference in the dogs' retrieval performance between the direct and overhearing conditions.

Researchers observed ($n = 10$) dogs and recorded the proportion of successful retrievals out of 12 tests for each dog. The observed proportions were

{1.0000, 1.0000, 0.7500, 1.0000, 0.4167, 0.8333, 0.7500, 0.8333, 0.8333, 1.0000}

They also considered the paired differences (direct – overhearing):

{−0.0833, 0.0000, −0.0833, −0.0833, 0.5833, 0.0833, −0.2500, 0.1667, 0.0000, −0.4167}

1.2.2 Hypotheses

We want to test if the dogs were better than chance, so we use right sided test:

$$H_0 : \mu_X = 0.50$$

$$H_a : \mu_X > 0.50$$

where μ_X is the population mean proportion of successful retrievals.

1.2.3 Conditions for Use

1. The variable of interest is quantitative.
2. The sample is probably representative of the larger population.
3. The sample observations are independent.

4. The underlying population distribution is not approximately Gaussian, since the observed data are left-skewed and have several observations at the upper bound of 1. In addition, the sample size ($n = 10$) is too small to use CLT.

1.2.4 One-Sample t-Test

The one-sample t-test was performed using the following R code:

```
1 x <- c(1.0000, 1.0000, 0.7500, 1.0000, 0.4167,
2       0.8333, 0.7500, 0.8333, 0.8333, 1.0000)
3
4 t.test(x, mu = 0.50, alternative = "greater")
```

One Sample t-test

```
data: x
t = 5.9386, df = 9, p-value = 0.0001092
alternative hypothesis: true mean is greater than 0.5
95 percent confidence interval:
 0.736198      Inf
sample estimates:
mean of x
 0.84166
```

The test statistic is calculated as

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The sample mean was $\bar{x} = 0.8417$. The test resulted in $t = 5.9386$ with $df = 9$ and a one-sided p-value of 0.0001092.

1.2.5 p-value

We use a right-tailed test ($\mu_X > \mu_0$).

- Observations of t that are substantially larger than 0.5, which correspond to observations of $\bar{x}$ that are substantially larger than μ_0 .
- $p\text{-value} = P(T \geq t) = P(T \geq 5.9386) = 0.0001092$

1.2.6 Results

We use the significance level $\alpha = 0.05$. Since the $p\text{-value} = 0.0001092$ is less than 0.05, we reject the null hypothesis. The data provide statistically discernible support for the alternative hypothesis.

The test statistic $t = 5.9386$ indicates that the observed sample mean is approximately 5.94 standard errors above the null hypothesized mean of 0.50. Therefore, the data suggest that the dogs' mean retrieval performance is better than chance.

1.2.7 Effect Size

To evaluate the magnitude of the effect, Hedges' g was calculated.

```
1 library(effectsize)
2
3 test.result <- t.test(x, mu = 0.50, alternative = "greater")
4
5 hedges_g(test.result)
```

```
Hedges' g |      95% CI
-----
```

1.72 | [0.88, Inf]

- Deviation from a difference of 0.5.
- One-sided CIs: upper bound fixed at [Inf].

The estimated effect size was Hedges' $g = 1.72$, indicating a large effect. The mean retrieval performance was approximately 1.72 standard deviations above the chance level of 0.50.

1.2.8 Confidence Interval

The 95% confidence interval was obtained from the t-test result:

```
1 test.result$conf.int
```

```
[1] 0.736198      Inf
attr(,"conf.level")
[1] 0.95
```

We are 95% confident that the population mean retrieval proportion is approximately $[0.736, \infty)$. Since this lower bound is above the chance level of 0.50, the confidence interval is consistent with the conclusion that the dogs performed better than chance.

2 Question 2

2.1 Part a

The data is qualitative, and each trial is a Bernoulli trial (either succeed or fail), so binomial distribution is possible. However, the trials may not be independent, since there should be correlations between each trial. Specifically, the performance of each dog in the two new-toy trials may tend to be consistent. Moreover, the probability of selecting the new toy by chance may not be exactly 50%, depending on dogs' behaviors.

2.2 Part b

The probability that a dog succeed in both trials are

$$p = (0.50)(0.50) = 0.25.$$

Let $X_i = 1$ be the dog i getting both toys right on the first try, and $X_i = 0$ be otherwise,

$$X_i \sim \text{Bernoulli}(0.25).$$

Therefore, the total number of dogs getting both toys right follows a binomial distribution.

$$B \sim \text{Binomial}(10, 0.25).$$

Since all 10 dogs got both trials right,

$$P(B = 10) = \binom{10}{10} (0.25)^{10} (0.75)^0 = (0.25)^{10} \approx 9.54 \times 10^{-7}.$$

Thus, the probability that all $n = 10$ dogs would get both trials correct is approximately 9.54×10^{-7} .

2.3 Part c

The Gaussian approximation has mean

$$\mu_Y = np = 10(0.25) = 2.5$$

and standard deviation

$$\sigma_Y = \sqrt{np(1-p)} = \sqrt{10(0.25)(0.75)} \approx 1.369.$$

Using the continuity correction,

$$P(B = 10) \approx P(9.5 < Y < 10.5).$$

And then we standardize it using

$$Z = \frac{Y - \mu_Y}{\sigma_Y}.$$

The lower and upper bounds are

$$\begin{aligned} z_{\text{lower}} &= \frac{9.5 - 2.5}{1.369} \approx 5.11, \\ z_{\text{upper}} &= \frac{10.5 - 2.5}{1.369} \approx 5.84 \end{aligned}$$

Therefore,

$$P(B = 10) \approx P(5.11 < Z < 5.84) \approx 1.57 \times 10^{-7}.$$

Compared to 9.54×10^{-7} , which is the binomial probability from Part b, the Gaussian probability is much smaller. They are not close.

This is because one of the conditions to apply Gaussian distribution to approximate probabilities, is to make sure that

$$np \geq 15, \text{ and } n(1-p) \geq 15.$$

In this case,

$$np = 10(0.25) = 2.5 \text{ and } n(1-p) = 10(1-0.25) = 7.5,$$

both of which are less than 15, so the Gaussian approximation conditions are not satisfied.

2.4 Part d

No, it doesn't make sense to make inference using the Gaussian method.

Given that all 10 dogs are successful, or $\hat{p} = 1$, the standard error of the sample would be 0, as calculated below:

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{1(0)}{10}} = 0.$$

When we try to estimate $Z = \frac{\text{estimate} - \text{nullvalue}}{SE}$, the denominator would become zero, which is undefined. It would be hard to estimate confidence interval too.